

8.333 Recitation 5
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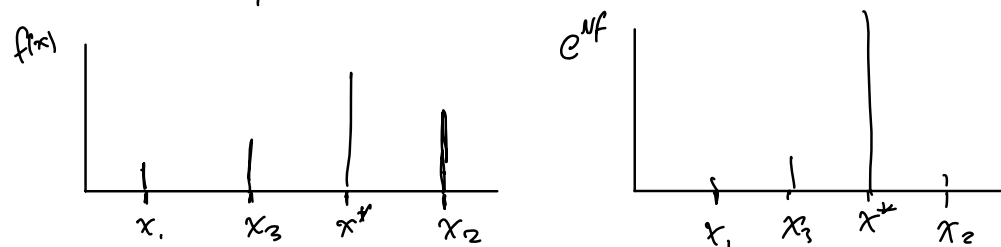
1 The Saddle-Point method

a) Discrete sums

Let $f: \mathbb{R} \rightarrow \mathbb{R}$, and consider a sequence $\{x_1, x_2, \dots\} \subseteq \mathbb{R}$. Suppose $f(x_i)$ has a unique maximum on this sequence $\max_i f(x_i) = f(x^*)$.

Consider the sum

$$I = \sum_i e^{Nf(x_i)}$$



As $N \rightarrow \infty$, the term $e^{Nf(x^*)}$ becomes exponentially larger than the rest:

$$\frac{e^{Nf(x^*)}}{e^{Nf(x_i)}} = e^{N(f(x^*) - f(x_i))}$$

So the sum is well approximated by

$$I \simeq e^{Nf(x^*)}$$

In the sense

$$\lim_{N \rightarrow \infty} \frac{I}{e^{Nf(x^*)}} = 1$$

corrections vanish exponentially

b) Continuous Integrals (Laplace's method)

Now suppose we instead have an integral over the real line

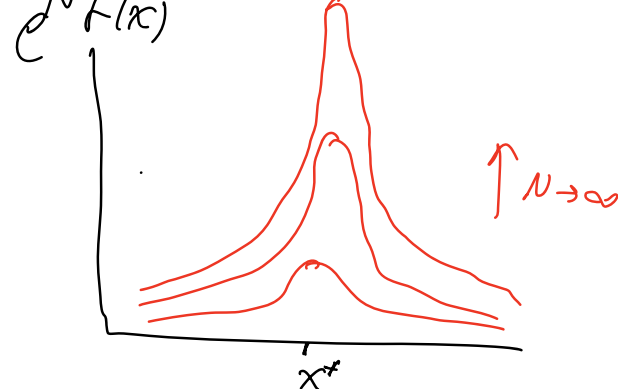
$$I = \int_a^b e^{Nf(x)} dx$$

Suppose again that f has a unique global maximum at x^* , $e^{Nf(x^*)}$, and let f be twice differentiable with $f''(x^*) \neq 0$. We again have

$$\frac{e^{Nf(x^*)}}{e^{Nf(x)}} = e^{N(f(x^*) - f(x))}$$

and so the integrand becomes sharply peaked at x^*

(example: let $f(x) = \frac{\sin x}{x}$)



Integral is dominated by behavior near x^* , so we Taylor expand:

$$f(x) = f(x^*) + \cancel{f'(x^*)} (x-x^*) + \frac{f''(x^*)}{2} (x-x^*)^2 + \sum_{k \geq 2} \frac{f^{(k)}(x^*)}{k!} (x-x^*)^k$$

Then

$$I = e^{Nf(x^*)} \int dx \exp \left[N \frac{f''(x^*)}{2} (x-x^*)^2 + O(N(x-x^*)^3) \right]$$

Gaussian integral:
($f''(x^*) < 0$)

$$I \simeq e^{Nf(x^*)} \sqrt{\frac{2\pi}{N|f''(x^*)|}} \quad \text{as } N \rightarrow \infty.$$

How do we show that higher order terms are subdominant? We will show

Theorem $\lim_{N \rightarrow \infty} \frac{I}{e^{Nf(x^*)} \sqrt{\frac{2\pi}{N|f''(x^*)|}}} = 1.$

Proof Def $R_N \equiv I e^{-Nf(x^*)} \sqrt{\frac{N|f''(x^*)|}{2\pi}}$

$$\begin{aligned} R_N &= \sqrt{\frac{N|f''(x^*)|}{2\pi}} \int dx \exp \left[N \frac{f''(x^*)}{2} (x-x^*)^2 \right] \exp \left[\sum_{k \geq 2} \frac{N f^{(k)}(x^*)}{k!} (x-x^*)^k \right] \\ &= \int dx p(x) \exp \left[\sum_{k \geq 2} \frac{N f^{(k)}(x^*)}{k!} (x-x^*)^k \right] \end{aligned}$$

where $p(x) \equiv \sqrt{\frac{N|f''(x^*)|}{2\pi}} \exp \left[N \frac{f''(x^*)}{2} (x-x^*)^2 \right]$ is a $\mathcal{N}(x^*, \frac{1}{N|f''(x^*)|})$ prob. density.

$$\begin{aligned} \therefore R_N &= \left\langle \exp \left[\sum_{k \geq 2} \frac{N f^{(k)}(x^*)}{k!} (x-x^*)^k \right] \right\rangle \quad \text{where } \langle \cdot \rangle \text{ is w.r.t } p(x) \\ &= \left\langle \sum_{m=0}^{\infty} \frac{N^m}{m!} \left(\sum_{k \geq 2} \frac{f^{(k)}(x^*)}{k!} (x-x^*)^k \right)^m \right\rangle \end{aligned}$$

$$= 1 + N \sum_{k \geq 2} \frac{f^{(k)}(x^*)}{k!} \langle (x - x^*)^k \rangle + \frac{N^2}{2} \sum_{\substack{k_1 \geq 2 \\ k_2 \geq 2}} \frac{f^{(k_1)}(x^*) f^{(k_2)}(x^*)}{k_1! k_2!} \langle (x - x^*)^{k_1 + k_2} \rangle$$

$\nwarrow$ k^{th} central moment of Gaussian $\nwarrow$ $(k_1 + k_2)^{\text{th}}$ central moment of Gaussian

$$+ \frac{N^3}{6} \cdot (\dots) + \dots$$

All N -dependent terms have the form $N^m \langle (x - x^*)^{\sum_{i=1}^m k_i} \rangle$ with $\sum_{i=1}^m k_i \geq 2m$

Lemma The k^{th} central moment of a $\mathcal{N}(\mu, \sigma^2)$ distribution is:

$$\langle (x - \mu)^k \rangle = \begin{cases} 0 & \text{if } k \text{ odd} \\ a_k \sigma^k & \text{if } k \text{ even} \end{cases}$$

$\nearrow$ not important

where $a_k = (k-1)(k-3)\dots(2)$
 $= (k-1)!!$

$\nearrow$ obvious from Wick's theorem: $\langle x x \dots x \rangle = \sum b \underbrace{\langle x x \rangle \langle x x \rangle \dots \langle x x \rangle}_{k/2 \text{ times}} \propto \sigma^k$

Using this lemma, we see that $N^m \langle (x - x^*)^{\sum_{i=1}^m k_i} \rangle = 0$ if $\sum_{i=1}^m k_i$ is odd. Otherwise:

$$N^m \langle (x - x^*)^{\sum_{i=1}^m k_i} \rangle \propto N^m (N |f''(x^*)|)^{-\frac{1}{2} \sum_{i=1}^m k_i} \propto N^{m - \frac{1}{2} \sum_{i=1}^m k_i} \xrightarrow{N \rightarrow \infty} 0$$

This proves the theorem since $\sum_{i=1}^m k_i \geq 2m$.

$$\lim_{N \rightarrow \infty} R_N = 1$$



The calculation above gives us the scaling of the leading order correction:

$$R_N = 1 + N \frac{f^{(4)}(x^*)}{4!} \frac{3!!}{N^2 (|f''(x^*)|)^2} + \frac{N^2}{2} \frac{(f^{(3)}(x^*))^2}{3! 3!} \frac{5!!}{N^3 (|f''(x^*)|)^3} + \mathcal{O}(N^{-1})$$

$$= 1 + \left[\frac{1}{8} \frac{f^{(4)}(x^*)}{|f''(x^*)|^2} + \frac{5}{24} \frac{(f^{(3)}(x^*))^2}{|f''(x^*)|^3} \right] \frac{1}{N} + \mathcal{O}(N^{-1})$$

Thus

$$\mathcal{I} \simeq e^{N f(x^*)} \sqrt{\frac{2\pi}{N |f''(x^*)|}} (1 + \mathcal{O}(1/N))$$

Proof of lemma:

$$\begin{aligned}\langle (x-\mu)^k \rangle &= \frac{1}{\sqrt{2\pi\sigma^2}} \int dx (x-\mu)^k e^{-\frac{(x-\mu)^2}{2\sigma^2}} \\ &= \frac{\sigma^k}{\sqrt{2\pi}} \int dx x^k e^{-x^2/2} \quad x \rightarrow \frac{x-\mu}{\sigma}, \quad dx \rightarrow \sigma dx \\ &\quad (x-\mu)^k \rightarrow \sigma^k x^k \\ &\equiv a_k \sigma^k\end{aligned}$$

$$\begin{aligned}a_k &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dx x^k e^{-x^2/2} \\ &= \frac{2^{k/2}}{\sqrt{\pi}} \int_0^{\infty} dt t^{(k-1)/2} e^{-t} \\ &= \frac{2^{k/2}}{\sqrt{\pi}} \Gamma\left(\frac{k}{2} + \frac{1}{2}\right) \\ &= \frac{2^{k/2}}{\sqrt{\pi}} \frac{(k-1)!! \sqrt{\pi}}{2^{k/2}} \\ &= (k-1)!!\end{aligned}$$

$$t = \frac{x^2}{2}, \quad dt = x dx$$

$$x^k = (2t)^{k/2}$$

$$dx x^k = \frac{dt}{x} x^k = dt (2t)^{(k-1)/2}$$

See Artken 1985 p 548



c) Stirling's approximation

We can apply the saddle point method to derive Stirling's approximation and its corrections.

A silly way to get the leading logarithmic behavior is:

$$\log n! = \sum_{k=1}^n \log k \approx \int_1^n dx \log x = n \log n - n + 1$$

Saddle point method gives us the full formula and a way to compute finite N corrections:

$$N! \equiv \Gamma(N+1) = \int_0^{\infty} dx x^N e^{-x} = \int_0^{\infty} dx e^{N \log x - x}$$

Change var $y = x/N$

$$N! = N \int_0^{\infty} dy e^{N \log(Ny) - Ny} = N e^{N \log N} \int_0^{\infty} dy e^{N(\log y - y)}$$

As $N \rightarrow \infty$, use Laplace's method:

$$f(y) = \log y - y$$

$$\text{Maximized at } \frac{1}{y} - 1 = 0 \rightarrow y^* = 1$$

$$f''(y) = -y^2 \Rightarrow f''(y^*) = -1$$

$$\text{So } N! \simeq N e^{N \log N} e^{-N} \sqrt{\frac{2\pi}{N}} = \boxed{\sqrt{2\pi N} \left(\frac{N}{e}\right)^N}$$

How about the first correction?

$$\left[\frac{1}{8} \frac{f^{(4)}(x^*)}{|f''(x^*)|^2} + \frac{5}{24} \frac{(f^{(3)}(x^*))^2}{|f''(x^*)|^3} \right] = \frac{1}{8} \frac{(-6)}{1} + \frac{5}{24} \frac{2^2}{1} = \frac{1}{12}$$

$$\therefore \boxed{N! = \sqrt{2\pi N} \left(\frac{N}{e}\right)^N \left[1 + \frac{1}{12N} + \mathcal{O}(N^{-2}) \right]}$$

d) Saddle Point method for probability distributions

Suppose we have a probability distribution of the form

$$p(x) \propto e^{N f(x)} \quad \text{called rate function in large deviation theory} \quad \text{Large deviation form}$$

with $\max f(x) = f(x^*)$, $f''(x^*) < 0$. For example, $P(E) = e^{\frac{N}{k}(S(E) + S(E-E_1))}$

The integral

$$I = \int dx e^{-N f(x)}$$

can be computed using the saddle point method (e.g. $\Omega(E) = \int dE_1 p(E_1)$).

i.e. Integral can be treated as a Gaussian of width $\sqrt{N}$. But can the distribution itself be treated as Gaussian? Yes! Consider the expectation value of any observable $G(x)$:

$$\langle G(x) \rangle = \frac{1}{I} \int dx G(x) e^{N f(x)}$$

Theorem If $G(x)$ admits a Taylor expansion about x^* , then

$$\lim_{N \rightarrow \infty} \frac{\langle G(x) \rangle}{\langle G(x) \rangle_{\text{Normal}}} = 1, \quad \text{where } \langle \cdot \rangle \text{ denotes an expectation wrt } P(x) \text{ and } \langle \cdot \rangle_{\text{Normal}} \text{ denotes an expectation wrt } \mathcal{N}\left(x^*, \frac{1}{N|f''(x^*)|}\right)$$

Proof Simple extension of the proof for the Laplace method:

$$\begin{aligned}\langle G(x) \rangle &= \frac{1}{I} \sum_k \frac{G^{(k)}(x^*)}{k!} \int dx (x-x^*)^k e^{Nf(x)} \\ &= \frac{e^{Nf(x^*)}}{I} \sum_k \frac{G^{(k)}(x^*)}{k!} \int dx \left(e^{\frac{Nf''(x^*)}{2}(x-x^*)^2} \right) (x-x^*)^k \left[\exp\left(\sum_{l \geq 2} \frac{Nf^{(l)}(x^*)}{l!} (x-x^*)^l \right) \right]\end{aligned}$$

We see that we can carry out the same proof as before, because the factors $(x-x^*)^k$ come with no N dependence and thus vanish as N^{-k} :

The result above implies that the moments of $p \propto e^{Nf(x)}$ approach those of a $\mathcal{N}\left(x^*, \frac{1}{N|f''(x^*)|}\right)$ distribution. 

Remark The requirement of analyticity at x^* is restrictive: suppose

$\mathcal{O}(x) = \Theta(x-a)$ with $a > x^*$. Then $\langle \mathcal{O} \rangle = \mathbb{P}(x > a)$ is a tail probability which is not well approximated by a Gaussian. This is the Large Deviation Regime (next recitation).

2) Discrete Stat Mech

Everything we have done in class so far has been with continuous phase spaces. With discrete state spaces, life is easy:

$$\begin{aligned}\Omega(E) &= \text{number of configurations with energy } E \\ &= |\Gamma(E)|\end{aligned}$$

where $\Gamma(E)$ is the space of all configurations with energy E .

The probability measure is uniform:

$$\mathbb{P}[w] = \frac{1}{\Omega(E)}$$

Probability of a macrostate $M \subseteq \Gamma(E)$ is thus:

$$\mathbb{P}[w \in M] = \sum_{w \in M} \frac{1}{\Omega(E)} = \frac{|M|}{\Omega(E)}$$

There are no dimensional issues with the entropy: $[\Omega] = 1$, so $S = k \log \Omega$ is fine. No need for ΔE .

Two level Systems

As an example, consider a collection of N independent spins with energy ϵ when pointing up and 0 when pointing down: $H = \sum_{i=1}^N \epsilon_i$
 $\epsilon_i \in \{0, \epsilon\}$

$$\Omega(E = n\epsilon) = \binom{N}{n} = \frac{N!}{(\frac{E}{\epsilon})! (N - \frac{E}{\epsilon})!}$$

Entropy:

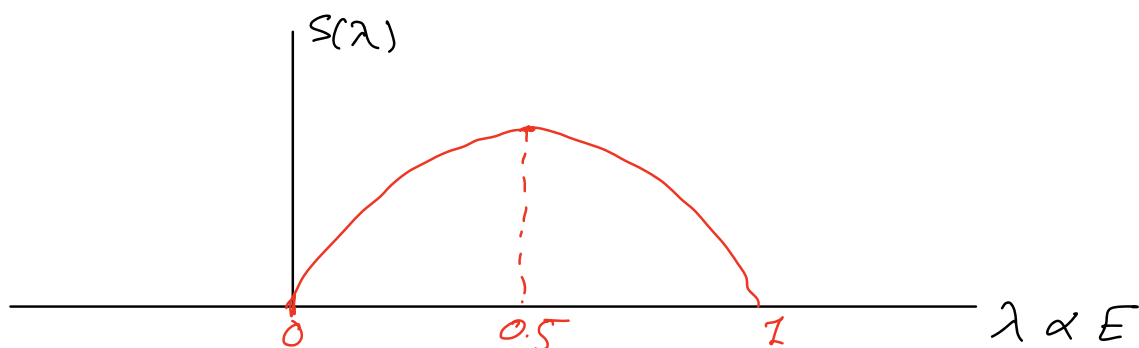
$$S = k \log \Omega \approx kN(\log N - 1) - \frac{kE}{\epsilon} \left[\log \frac{E}{\epsilon} - 1 \right] - k(N - \frac{E}{\epsilon}) \left[\log(N - \frac{E}{\epsilon}) - 1 \right]$$

$$= -Nk [\lambda \log \lambda + (1 - \lambda) \log(1 - \lambda)]$$

where $\lambda \equiv E / N\epsilon$

Notice $S = 0$ for $\lambda = 1$ or 0, because these cases correspond to all spins up or down and have $\Omega = 1$.

$\Rightarrow S$ is not monotonic in E



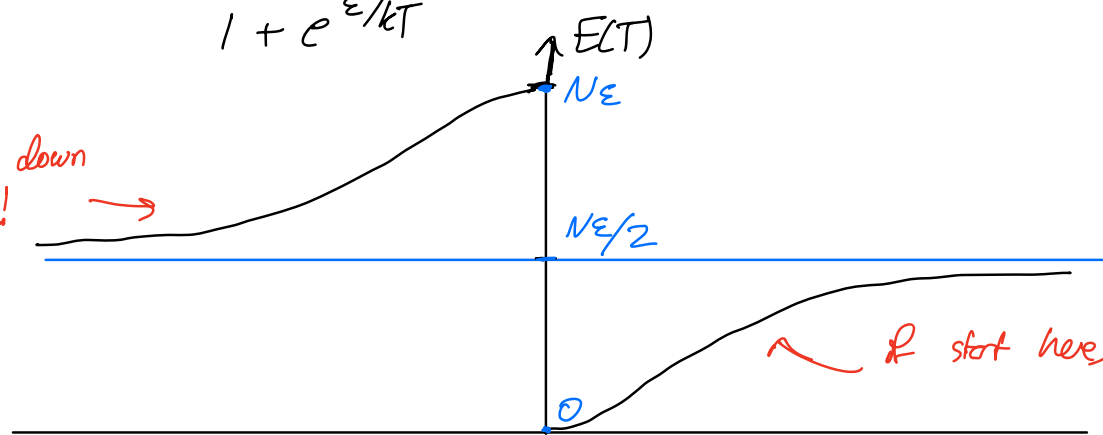
$$\frac{1}{T} = \frac{\partial S}{\partial E} = \frac{1}{N\epsilon} \frac{\partial S}{\partial \lambda} = \frac{k}{\epsilon} \log\left(\frac{1-\lambda}{\lambda}\right) < 0 \text{ if } \lambda > \frac{1}{2}!$$

Can invert this:

Negative temperature!

$$E(T) = N\epsilon \lambda = \frac{N\epsilon}{1 + e^{\epsilon/kT}}$$

Cool system down
by heating it! $\rightarrow$



This is due to the upper bound on the energy.

Consider now the heat capacity

$$C_V = \frac{dE}{dT} = Nk \frac{\epsilon^2}{(kT)^2} \frac{e^{\epsilon/kT}}{(1 + e^{\epsilon/kT})^2}$$

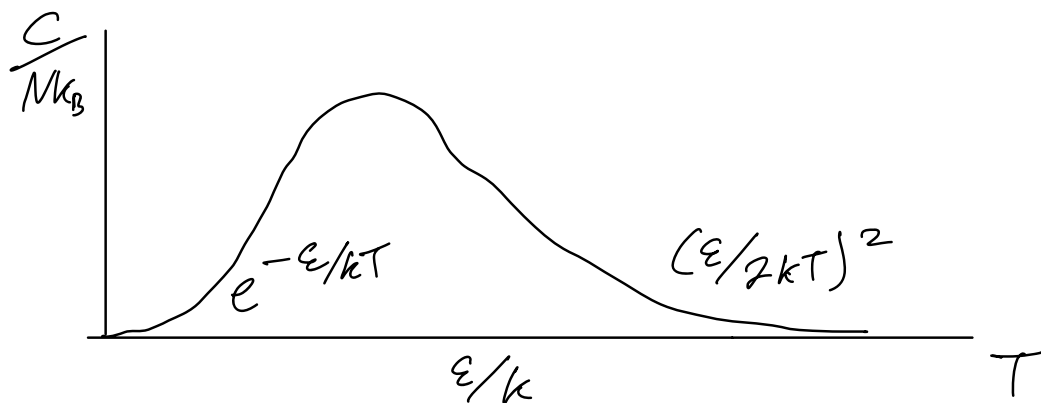
As $T \rightarrow 0$,

$$C_V \sim \frac{Nk\epsilon^2}{(kT)^2} e^{-\epsilon/kT} \Rightarrow \text{heat capacity vanishes exponentially as } T \rightarrow 0.$$

This is typical of systems with an energy gap above the ground state.

As $T \rightarrow \infty$

$C_V \rightarrow 0 \Rightarrow$ saturation effect because the system has a max energy.



The microcanonical ensemble is a full joint probability. What is the probability that spin 1 is pointing down? Marginalize the remaining spins:

$$P(\epsilon_1 = 0) = \frac{1}{\Omega(E, N)} \cdot (\text{number of configurations with energy } E \text{ and } \epsilon_1 = 0)$$

$$= \frac{\Omega(E, N-1)}{\Omega(E, N)} = \frac{\binom{N-1}{n}}{\binom{N}{n}} \quad \text{where } n \equiv E/\epsilon$$

$$= \left(\frac{(N-1)!}{n!(N-n-1)!} \right) \left(\frac{n!(N-n)!}{N!} \right) = \frac{N-n}{N}$$

$$= 1 - \frac{E}{n\epsilon} = 1 - \frac{1}{1 + e^{\epsilon/kT}}$$

$$\therefore P(\epsilon_1 = 0) = \frac{1}{1 + e^{\epsilon/kT}}, \quad P(\epsilon_1 = \epsilon) = \frac{e^{-\epsilon/kT}}{1 + e^{-\epsilon/kT}}$$

We will see that this is the answer one would get in the canonical ensemble $\Rightarrow$ simple example of ensemble equivalence.